

Probability and Statistics (MT-2005)

Homework # 03(c)

Chebyshev Theorem & Moment Generating Function

1. Let X and Y be random variables with finite expectations. Prove that:

- (a) $Var(aX + bY + c) = a^2Var(X) + b^2Var(Y) + 2abCov(X, Y)$
- (b) $Cov(aX + b, cY + d) = acCov(X, Y)$
- (c) $Corr(aX + b, cY + d) = Corr(X, Y)$

2. Let X and Y be random variables with finite expectations. Prove that:

$$-1 \leq Corr(X, Y) \leq 1$$

(Hint: Use Cauchy Schwarz Inequality $(E(XY))^2 \leq E(X^2)E(Y^2)$.)

3. A random variable X has mean $\mu = 100$ and variance $\sigma^2 = 25$. Use Chebyshev's inequality to find an upper bound on:

- (a) $P(|X - 100| \geq 10)$
- (b) $P(|X - 100| \geq 20)$
- (c) $P(50 < X < 150)$
- (d) Find c such that $P(|X - 100| \geq c) \leq 0.05$

4. A battery manufacturer claims that its AA batteries have a mean lifespan of 1200 hours with a standard deviation of 80 hours. The packaging states that the batteries last at least 1000 hours. Using Chebyshev's inequality, determine at most what percentage of the batteries fail before 1000 hours. Assume the lifespan distribution is symmetric about the mean.

5. In a support system in the U.S. space program, a single crucial component works only 85% of the time. In order to enhance the reliability of the system, it is decided that 3 components will be installed in parallel such that the system fails only if they all fail. Assume that the components act independently and that they are equivalent in the sense that all 3 of them have an 85% success rate. Consider the random variable X as the number of components out of 3 that fail.

- (a) Write a probability function for the random variable X .
- (b) Construct the moment-generating function for the random variable X .
- (c) Find the first cumulant κ_1 and second cumulant κ_2 of the given probability distribution.
- (d) Verify $\kappa_1 = E(X)$ and $\kappa_2 = Var(X)$